

Blackjack Monte Carlo

Strategy & Betting Analysis

74 Million Simulated Hands • 4 Action Strategies • 9 Betting Configurations

Part of AI Journey — Phase 2: Data & ML Engineering

github.com/arda-basarici/ai-journey

Executive Summary

This report presents findings from a large-scale Monte Carlo simulation of Blackjack, built to answer two questions: how much does playing strategy matter, and what do betting systems actually do to long-term outcomes?

The simulator was designed from scratch with a pluggable architecture — any strategy, from a simple rule set to a neural network, implements a single interface and plugs in without touching the simulation engine. This makes it a reusable research platform, not a one-time analysis tool.

Finding	Result
Basic strategy vs random play	House edge: 0.76% vs 42.4% — a 41.6 point difference
Lower bust rate = better strategy?	No. Semi-Random busts less than Basic but loses 8x more
Martingale betting system	40.9% of sessions end in complete ruin. Expected value unchanged
Bet variation without information	Identical results to flat betting — information is the edge
Card counting (6 decks, HiLo)	Net loss reduced from -\$69 to -\$8 per 1,000-hand session
Card counting (single deck, full)	+\$34 net profit per session — genuine player edge achieved
Risk of ruin driver	Bet sizing, not decision quality. Martingale: 40.9%, Flat: 0.6%

1. Strategy Matters — But Not How You Think

Four strategies were simulated across 1,000,000 hands each under standard Las Vegas Strip rules: 6 decks, dealer hits soft 17, blackjack pays 3:2. The strategies range from mathematically optimal (Basic) to completely random — spanning the full spectrum of possible play quality.

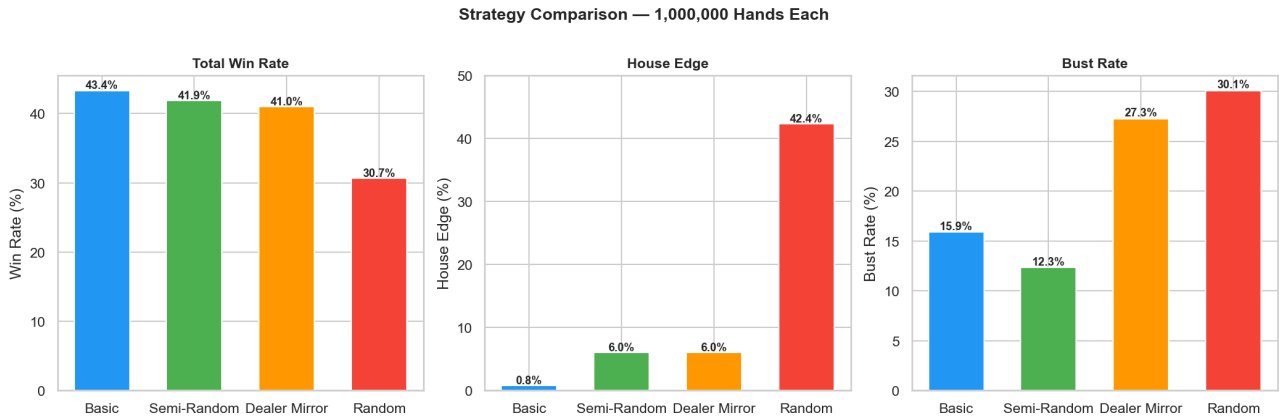


Figure 1: Win rate, house edge, and bust rate across four strategies (1M hands each).

Basic strategy reduces house edge from 42.4% to 0.76% — a 41.6 percentage point improvement over random play. This is the known mathematical optimum for Vegas Strip rules, and the simulation validates it precisely.

The Counterintuitive Finding

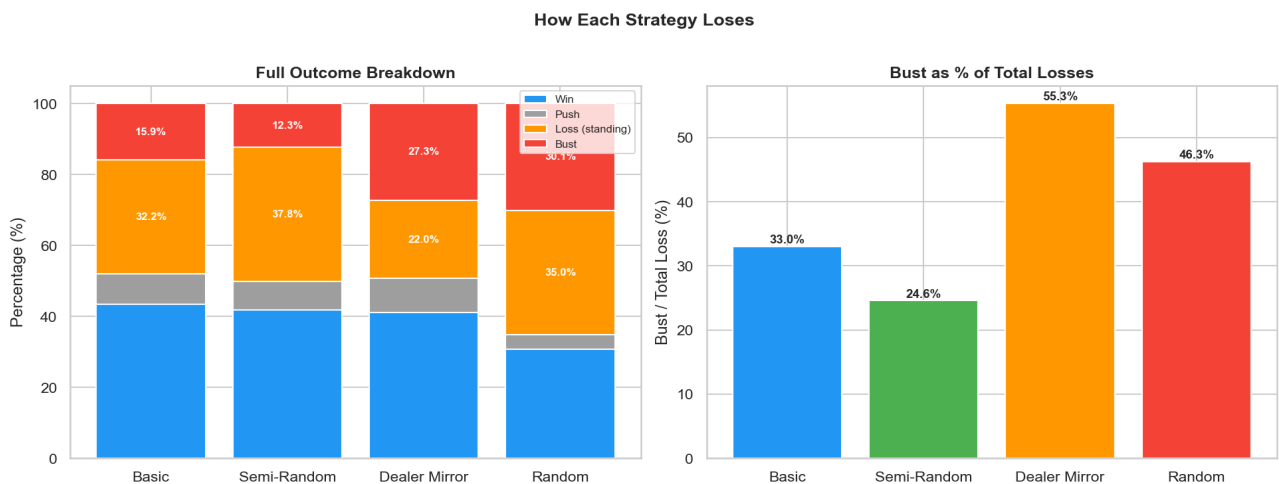


Figure 2: How each strategy loses — bust vs standing loss breakdown.

Semi-Random has the lowest bust rate of all four strategies at 12.3% — lower even than Basic Strategy's 15.9%. Yet its house edge is 6.03%, nearly 8 times worse than Basic's 0.76%. This directly contradicts the common intuition that avoiding busts is the primary goal in Blackjack.

Semi-Random randomly stands on hard 12-16. This avoids busts — but standing on 13 vs dealer 10 is a losing play regardless. The dealer completes a strong hand most of the time, and 13 loses to all of them. By randomly standing, Semi-Random trades a bust loss for a standing loss. The outcome is the same: a loss. But the bust counter goes down. Basic strategy hits in these situations because the alternative is worse, accepting calculated bust risk in exchange for better expected value.

Avoiding busts is not the goal. Maximizing expected value is.

2. Where the Edge Comes From

The win rate heatmap shows the objective difficulty of each game state — independent of what decision the player makes next. Green indicates favorable situations, red indicates unfavorable ones.

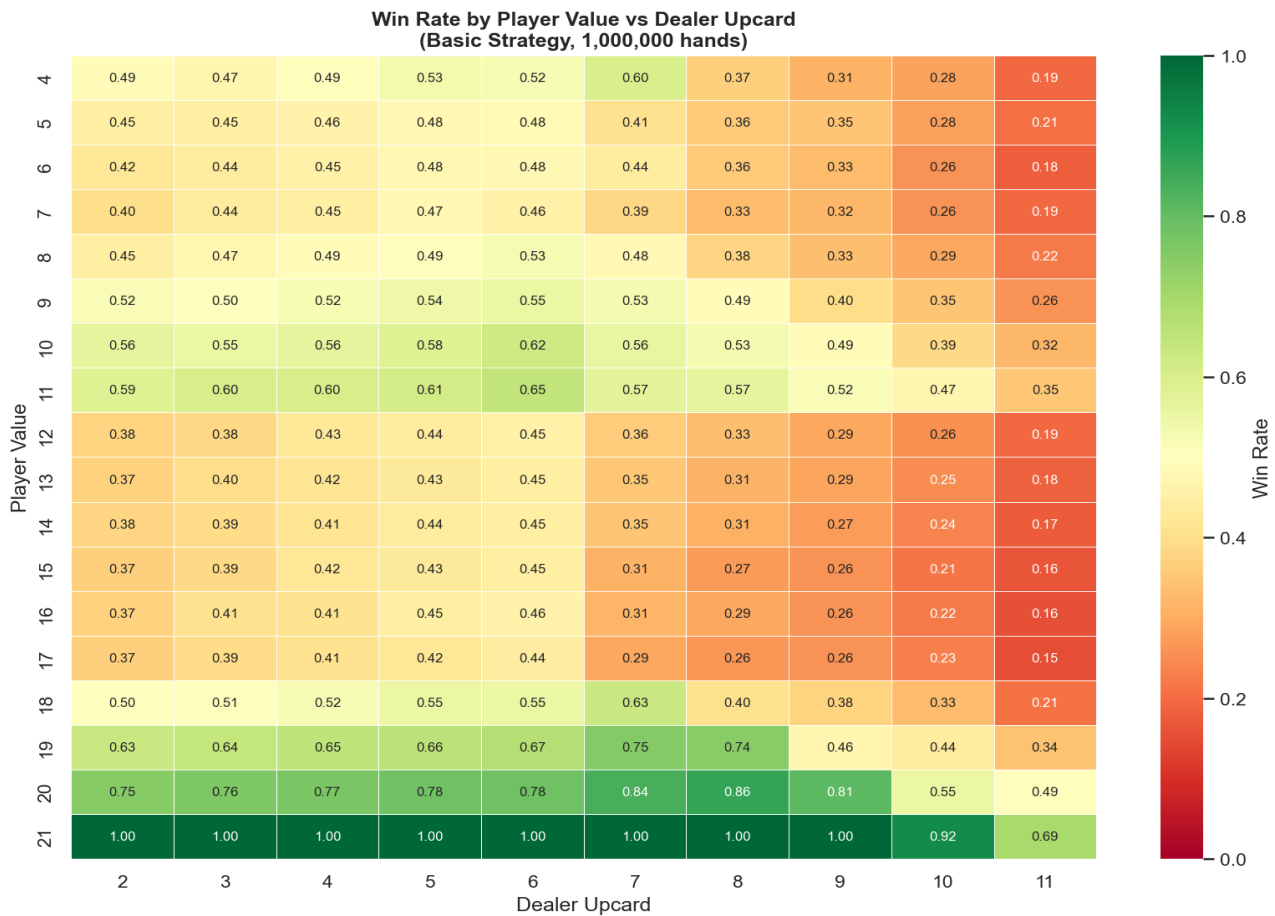


Figure 3: Win rate by player value vs dealer upcard — Basic Strategy, 1,000,000 hands.

The heatmap reveals the structural reality of the game: most states are red or orange. The player acts first, risking a bust before the dealer plays. This structural disadvantage cannot be eliminated — only managed.

Three patterns stand out. Dealer upcards 2-6 are noticeably greener — the dealer is likely to bust, making even weak player hands competitive. Dealer 10-11 columns shift deep red — the dealer rarely busts and completes strong hands frequently. Player values 12-16 vs dealer 7+ form the danger zone: red across the board, no good option, only a least-bad one. This is where strategy matters most, and where Basic Strategy's edge over naive approaches is concentrated.

Basic strategy's advantage comes from three specific situations: doubling on 9-11 vs weak dealer upcards (extracting maximum value from favorable positions), standing on 12-16 vs dealer 2-6 (letting a likely-busting dealer self-destruct rather than risking a premature bust), and splitting pairs correctly. Dealer Mirror and Semi-Random miss all of these. The win rate difference on hard 16 vs dealer 5-6 alone is up to 18 percentage points.

3. Your Money Over Time

Hand analysis tells you which decisions are correct. Session analysis tells you what happens to your money. 10,000 sessions of 1,000 hands each were simulated — 10 million hands per configuration — to map the long-term reality of each strategy and betting combination.

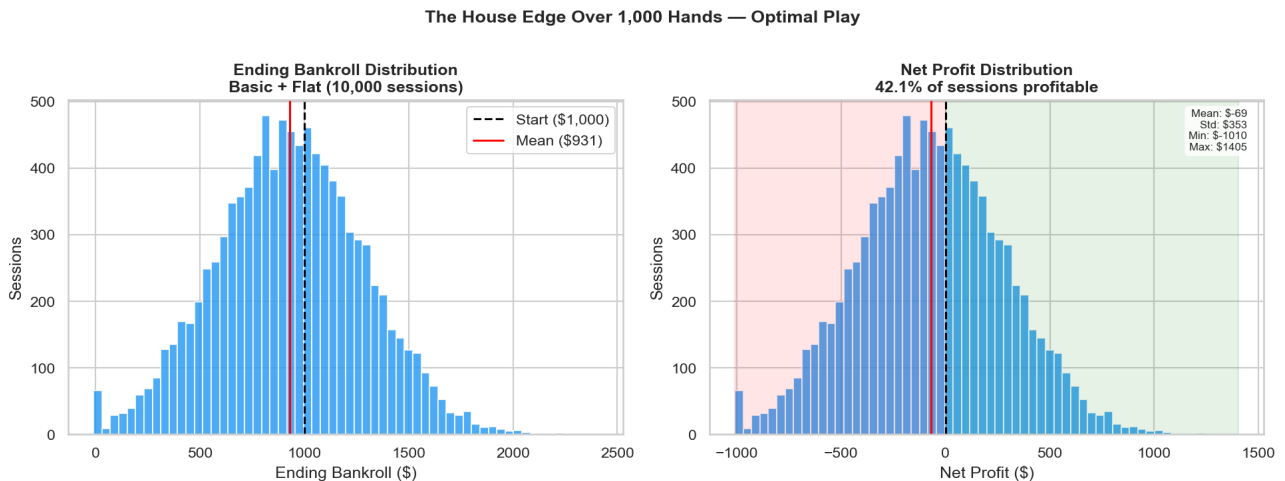


Figure 4: Ending bankroll and net profit distribution — Basic Strategy + Flat Betting, 10,000 sessions of 1,000 hands, starting bankroll \$1,000.

Even with mathematically optimal play, the average session ends with a \$69 loss on a \$1,000 starting bankroll. This is the house edge of 0.76% applied across 1,000 hands at \$10 per hand — expected and correct.

But variance is enormous. The standard deviation of ending bankroll is over \$350. Some sessions end up \$500+, others down \$600+. Both are normal outcomes of the same strategy. Critically, 42% of sessions end profitable despite a negative expected value. This has a direct practical implication: short-term results are not evidence of strategy quality. A player who wins \$300 in one session and concludes their system works, or loses \$400 and concludes basic strategy is wrong, is reading noise as signal. Only aggregate results across hundreds of sessions reveal the true edge.

4. The Betting System Myth

Progressive betting systems — Martingale in particular — are among the most persistent myths in gambling. The theory sounds compelling: double after every loss, and a single win recovers all previous losses plus one unit profit. The data tells a different story.

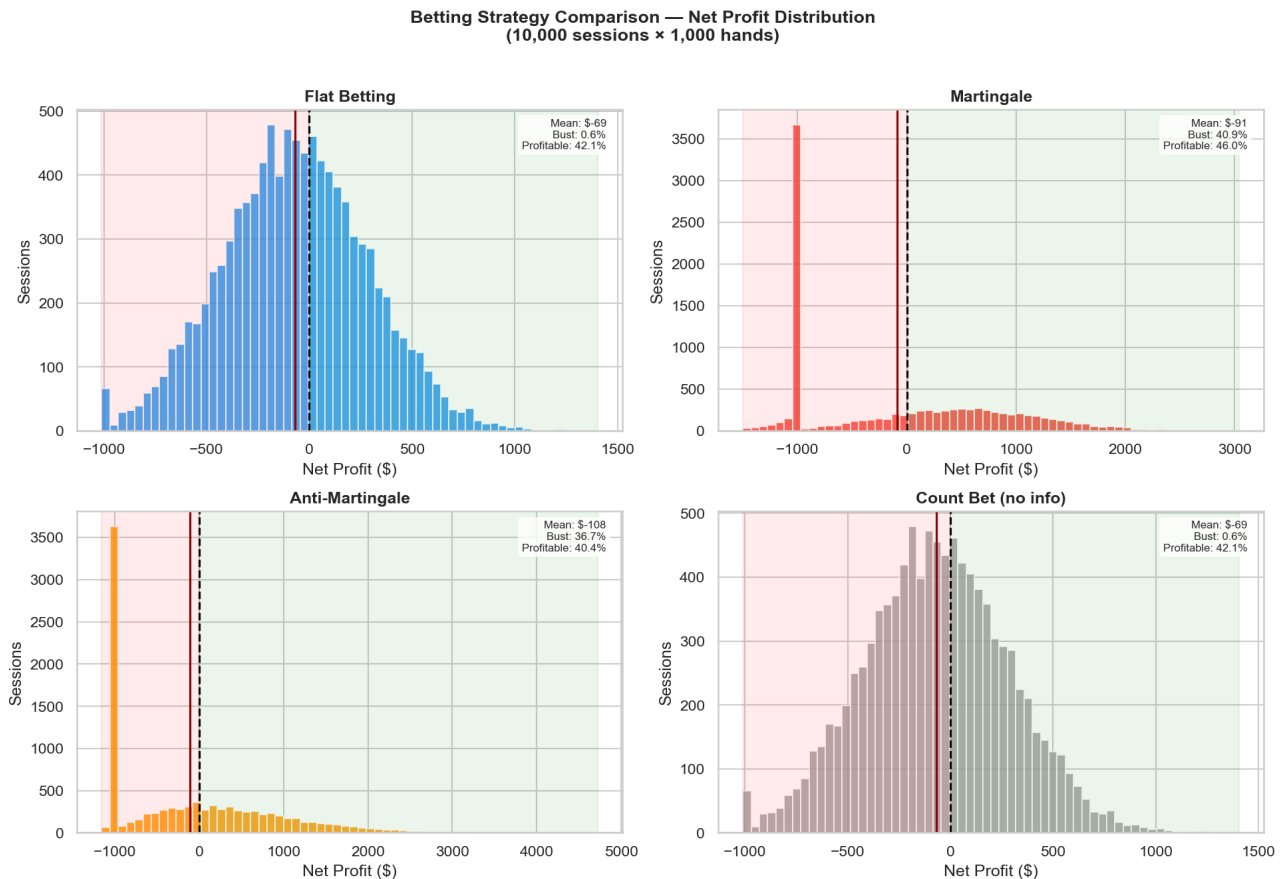


Figure 5: Net profit distribution across four betting strategies — Basic Strategy, 10,000 sessions of 1,000 hands each.

Martingale: The Numbers

Bust rate: 40.9%. Nearly half of all 1,000-hand sessions end in complete ruin. Average net profit: -\$91, worse than flat betting's -\$69. The Martingale distribution is bimodal — either the player survives and ends near breakeven, or a losing streak exceeds the bankroll before recovery is possible.

The mathematics is unchanged. Martingale does not alter the house edge. It converts small frequent losses into rare catastrophic ones. In real casinos, table limits prevent recovery from just 8-10 consecutive losses — making Martingale even more dangerous in practice than in this simulation, which has no table limits.

The Control Experiment

Count-based betting without a counting system produces results identical to flat betting: mean -\$69, bust rate 0.6%, indistinguishable distribution. Varying bet sizes without information does nothing. The information is the edge — not the betting pattern. This directly refutes the belief that varying bets creates any advantage.

Betting systems do not change expected value. They change the distribution of outcomes — adding variance and bust risk while leaving the house edge unchanged.

5. Card Counting — Reality vs Myth

Card counting is simultaneously overestimated and misunderstood. Hollywood portrays it as a guaranteed path to riches. Casinos treat it as cheating. The reality is more nuanced: it is a legitimate information-based technique that, under the right conditions, can create a genuine player edge — but the conditions matter enormously.

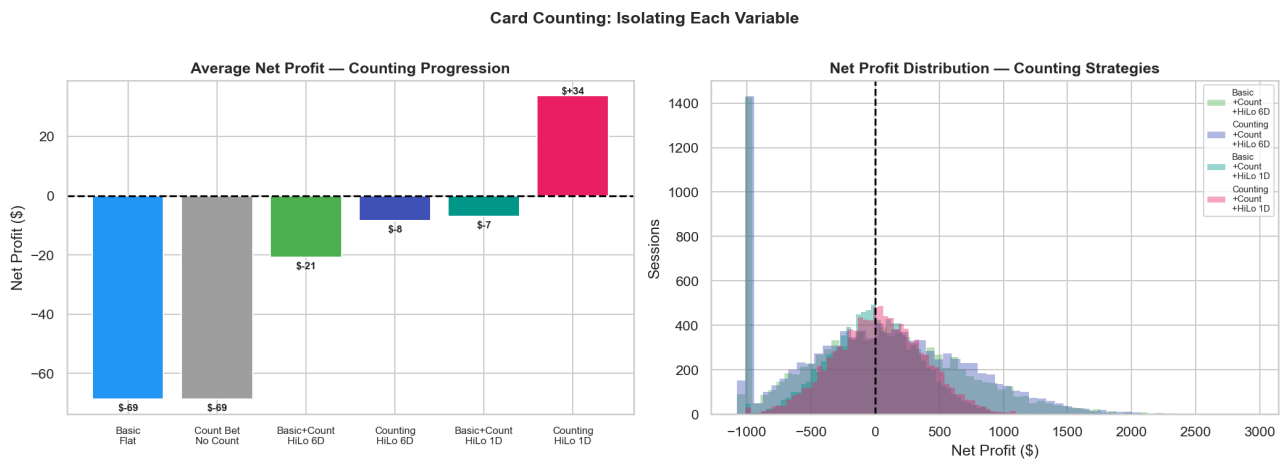


Figure 6: Average net profit per 1,000-hand session — isolating each component of card counting. All runs: 10,000 sessions, \$1,000 starting bankroll, \$10 base bet.

The progression isolates each variable. Bet variation alone (count-based betting without a counting system) produces zero effect — identical to flat betting. Adding HiLo counting to bet variation in a 6-deck game reduces net loss from \$69 to \$21 — a meaningful \$48 improvement. Adding index plays (strategy deviations based on count) reduces losses further to \$8. Switching to a single-deck game with full counting flips the edge to +\$34 per session.

The single-deck result of +\$34 average net profit across 10,000 sessions is a genuine player edge — not variance. The distribution shifts visibly rightward. Card counting works on single deck because each card dealt has a larger impact on deck composition, count swings are larger and more frequent, and favorable situations arise more often. Modern casinos use 6-8 decks specifically to reduce this effect.

An important methodological note: ROI percentages are misleading when bet sizes vary. A flat bettor wagers \$10,000 over 1,000 hands. A count-based bettor may wager \$15,000-\$20,000 as bet sizes scale with favorable counts. All comparisons here use net profit in dollars — the only fair comparison when denominators differ.

6. Risk of Ruin

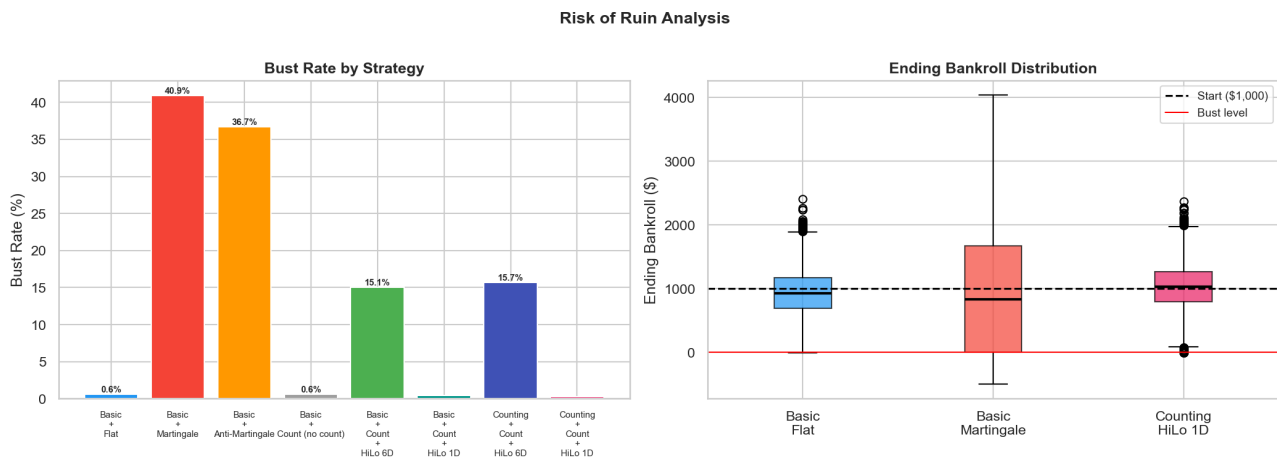


Figure 7: Bust rate by strategy configuration and ending bankroll distribution for three representative strategies.

Risk of ruin varies dramatically across configurations — from 0.3% to 40.9% — but is largely independent of decision quality. A player using perfect basic strategy with Martingale betting goes broke 40.9% of sessions. The same player with flat betting busts 0.6% of the time. The playing decisions are identical. Only the betting strategy differs.

The box plot shows the contrast starkly: flat betting's distribution stays well above zero in almost all sessions, while Martingale produces a long lower tail reaching zero — catastrophic sessions that end in complete ruin. The counting strategy on single deck achieves the best combination: positive expected value and a bust rate of only 0.3%.

Bet sizing and bankroll management determine survival. Playing strategy determines the rate of loss or gain within that survival window. These are independent levers and must be considered separately.

7. What This Simulator Can Answer

The analyses presented here are a subset of what the simulator can investigate. Because every decision is recorded with full game context, new questions can be answered by running new simulations — no architectural changes required.

How much does the 6:5 payout rule cost?

The tough config (6:5) vs vegas (3:2) with identical strategy quantifies the single most impactful rule variation — estimated at 1.4 percentage points of house edge.

At what bet spread does counting become profitable in 6 decks?

Our 1-to-8 spread produces -\$8 net. A 1-to-12 or 1-to-16 spread would extract more value. The simulator can test this by modifying the CountBasedBetting class.

Does Omega II outperform Hi-Lo?

Both are implemented. A direct comparison would quantify whether the added complexity of a multi-level system produces measurable benefit.

What is the optimal bankroll for a given risk tolerance?

Risk of ruin as a function of starting bankroll and bet size — mapping the survival probability curve precisely.

Phase 3: Neural Network Strategy

The per-decision dataset generated by this simulator is the training data for Phase 3. A neural network will be trained on millions of (game_state, action, outcome) tuples to learn optimal strategy from data alone — without being told the rules of Blackjack. The key question: does the model rediscover Basic Strategy? Where does it deviate? Does it discover index plays independently?

The simulator plugs in the trained model as a strategy object with no changes to the simulation engine. The session analysis framework then evaluates it identically to all hand-coded strategies — directly comparable results across all metrics presented here.

Built as Part of AI Journey

A structured learning arc from Python foundations to AI engineering. Every project is real, complete, and publicly documented.

github.com/arda-basarici/ai-journey